
✧ Practical approach to ✧
Anti-platelet And Anticoagulant therapy

OUTLINE

- ▣ Atherosclerosis
- ▣ Anti-platelet
- ▣ Anticoagulant

01 ✨ Atherosclerosis ✨

Sub endothelium call

LDL → Oxidized LDL

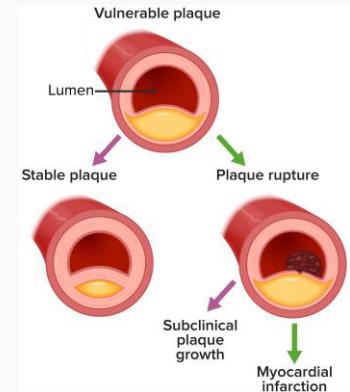
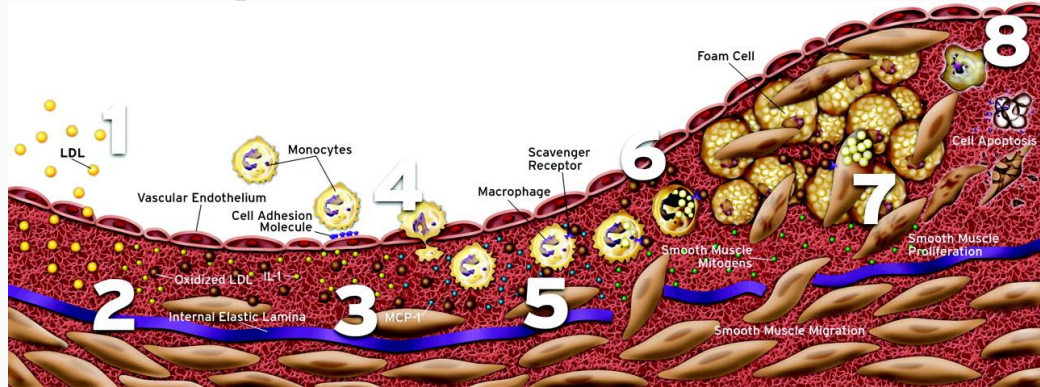
Macrophage ↓
Foam cell

Smooth muscle proliferation

Plaque

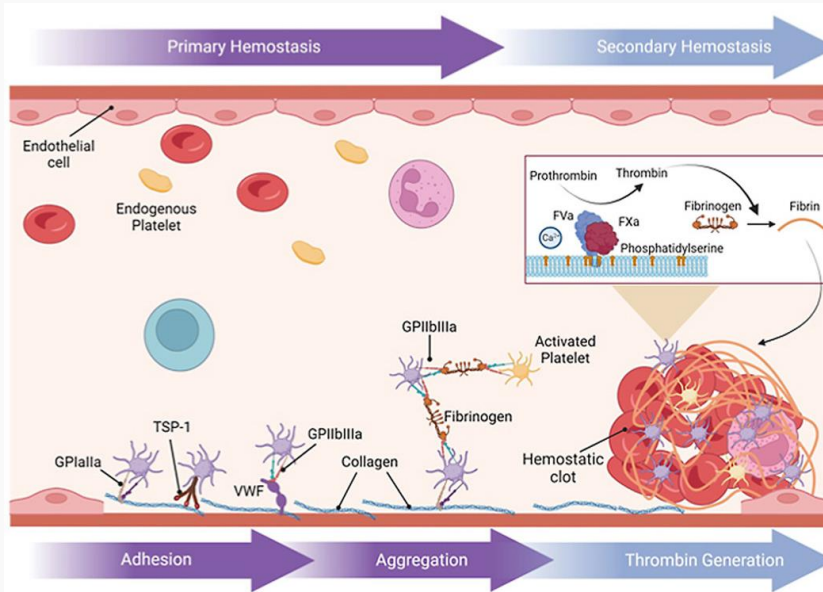
Stable plaque : Fibrous cap หนาแข็งแรง

Vulnerable plaque : Fibrous cap บาง
คลุ้มก่อนไขมันปริมาณมาก → Plaque rupture

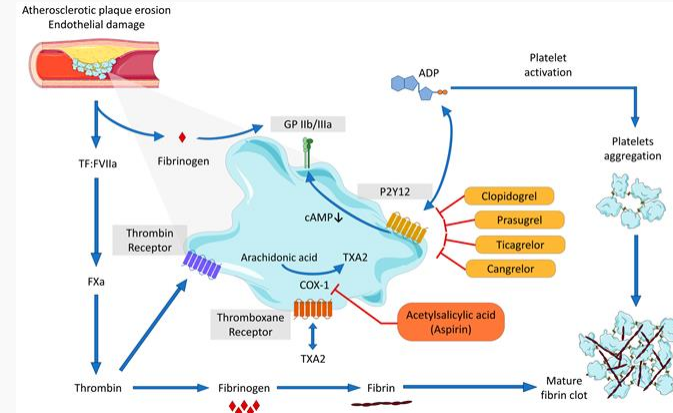


Vascular injury

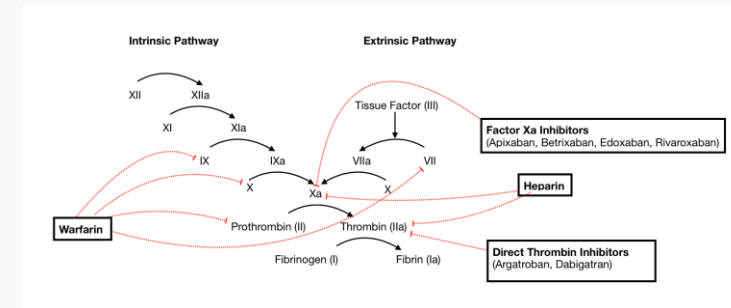
ระบบการแข็งตัวของเลือด

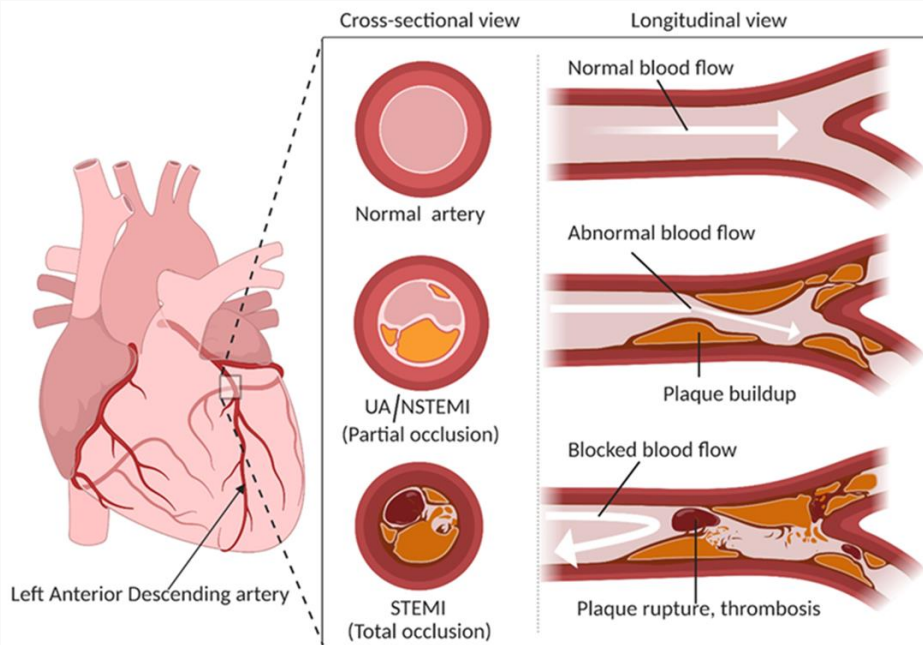


1.Primary Hemostasis แบบปฐมภูมิ



2.Secondary Hemostasis แบบทุติยภูมิ





CLINICAL PRACTICE GUIDELINES

2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines

Developed in Collaboration With and Endorsed by the American College of Emergency Physicians, National Association of EMS Physicians, and Society for Cardiovascular Angiography and Interventions



European Heart Journal (2023) 44, 3720–3826
<https://doi.org/10.1093/eurheartj/ehad191>

ESC GUIDELINES

2023 ESC Guidelines for the management of acute coronary syndromes

Developed by the task force on the management of acute coronary syndromes of the European Society of Cardiology (ESC)

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01 ✨ Anti-platelet ✨

1. Aspirin

❑ NSTEMI/STEMI

Loading dose : 162 – 325 mg oral

Maintenance : 75 - 100 mg oral once daily

Should be initiated in patients with ACS without contraindications as soon as possible



รูปแบบยา :

- ❑ Aspirin EC 300 mg
- ❑ Aspirin 81 mg

เคี้ยว **VS** ไม่เคี้ยว

EC : Enteric coated

สารเคลือบช่วยเพิ่มละลาย และการดูดซึม

2.P2Y₁₂ receptor inhibitor

Prodrug : inactive form

- Clopidogrel
- Prasugrel



CYP 2C19

Active form

- Ticagrelor : Active form

ระวังการใช้ยาคู่กับ

CYP 2C19 inhibitor

Fluoxetine

Fluconazole

Voriconazole

Omeprazole

Clopidogrel



❑ Binding site : ADP- Binding site : irreversible

Dose	NSTEMI	STEMI		
		Fibrinolytic		Without Fibrinolytic
		Age $\leq$ 75 ปี	Age $>$ 75 ปี	
Loading dose	300 mg	300 mg	-	600 mg
Maintenance dose	75 mg once daily			

Prasugrel

- ❑ Binding site : ADP- Binding site : irreversible.
- ❑ Contraindication : History stroke and TIA >>>> **Bleeding!!!**



Dose	NSTEMI	STEMI (Without Fibrinolytic)
	Undergoing PCI	
Loading dose	60 mg	
	Age ≥ 75 ȳ	Age < 75 ȳ
	Body weight < 60 kg	Body weight ≥ 60 kg
Maintenance dose	5 mg once daily	10 mg once daily

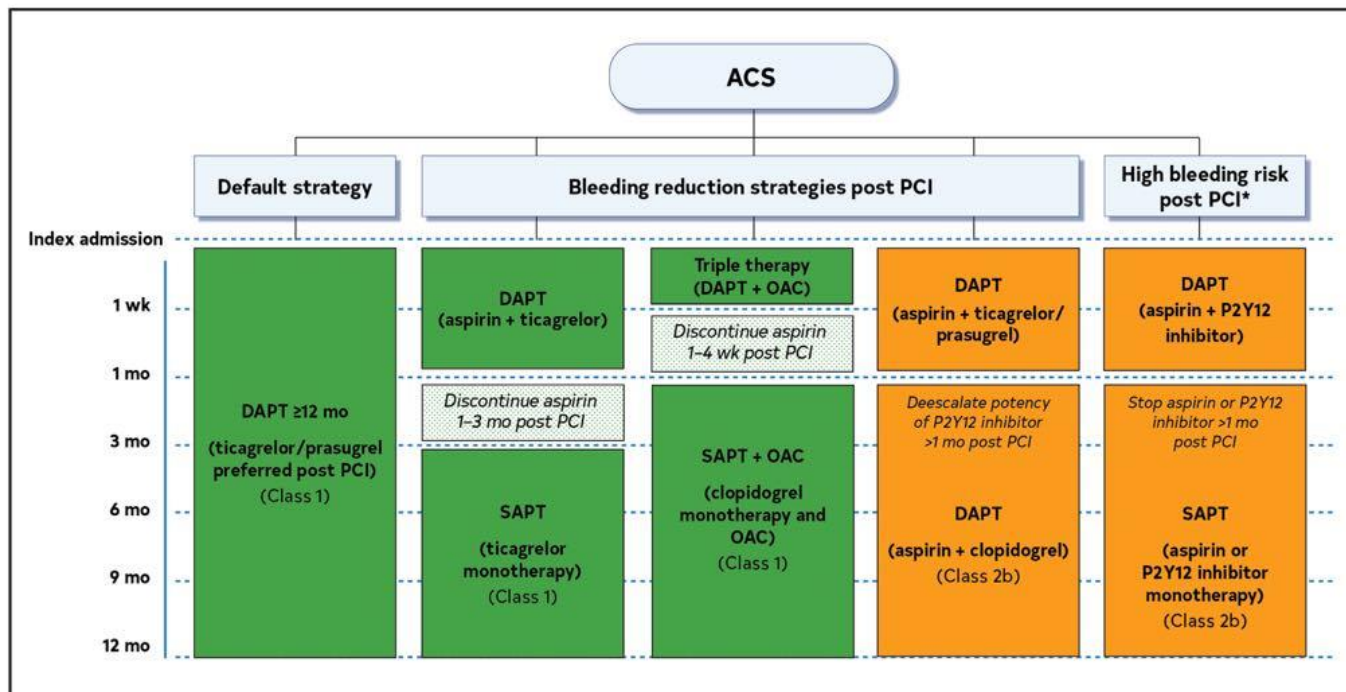
Ticagrelor

☐ Binding site : Allosteric binding site : reversible.

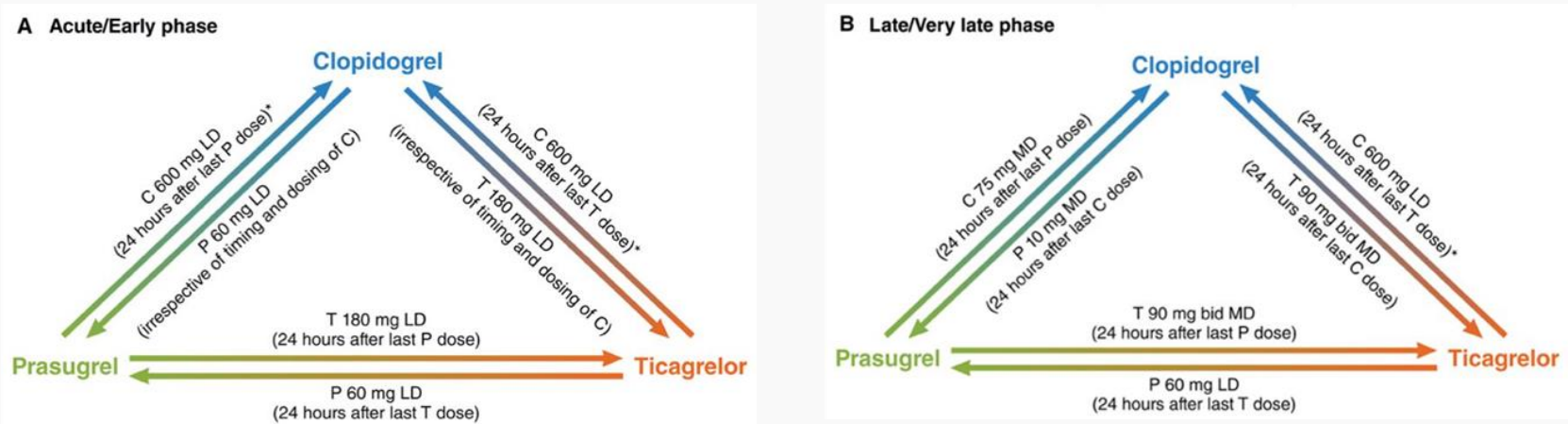


Dose	NSTEMI	STEMI (Without Fibrinolytic)
Loading dose	180 mg	
Maintenance dose	90 mg twice daily	

Duration



Switching between oral P2Y₁₂ receptor inhibitors



- (Acute phase) < 24 hr.
- (Early phase) 1- ≤ 30 day

- (Late phase) >30 day – 1 year
- (Very late phase) > 1 year

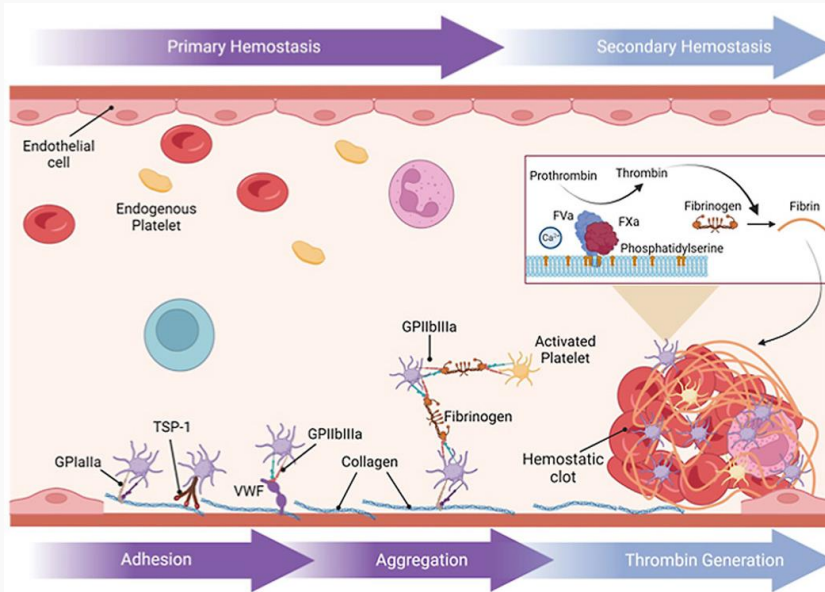
International expert consensus on switching platelet P2Y₁₂ receptor-inhibiting therapies. Circulation. 2017;136(20):1955–1975



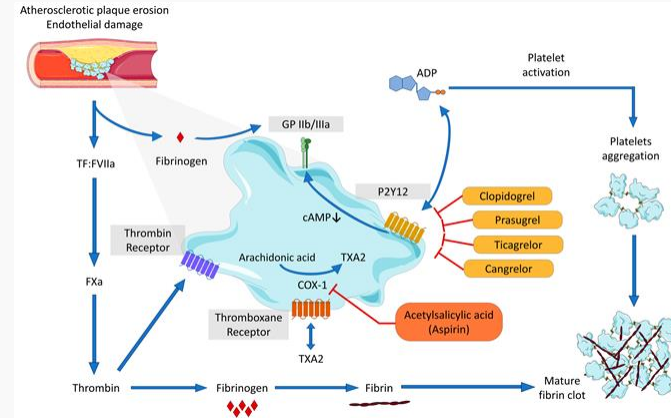
— Anticoagulant —

Vascular injury

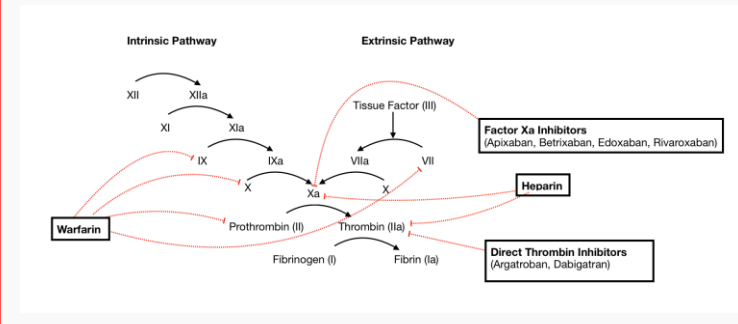
ระบบการแข็งตัวของเลือด



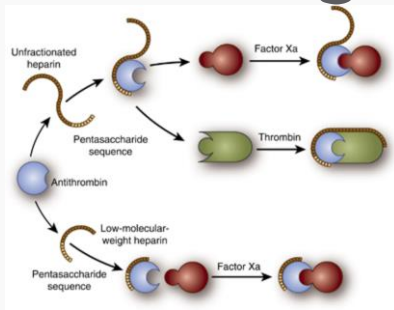
1.Primary Hemostasis แบบปฐมภูมิ



2.Secondary Hemostasis แบบทุติภูมิ



Anticoagulant : Parenteral



	Unfractionated Heparin	Low Molecular weight Heparin Enoxaparin	Fondaparinux
Mechanism of action	ยับยั้ง Factor Xa และ Thrombin (IIa)	ยับยั้ง Factor Xa > Thrombin (IIa)	ยับยั้ง Factor Xa (selective)
Indication	-Acute coronary syndrome -Venous thromboembolism		

Unfractionated Heparin : Dose



Acute coronary syndrome

- ☐ Initial therapy: IV bolus 60 IU/kg (max 4,000 IU)
then Initial infusion 12 unit/kg/hr.
(max 1,000 IU/hr.)

- ☐ To Support PCI : IV bolus 70-100 IU/kg

Venous thromboembolism

- ☐ DVT/PE

: IV bolus 80 IU/kg (max 5,000 IU) then Initial infusion 18 unit/kg/hr.

Monitoring : aPTT ratio หลังให้ยา 6 ชั่วโมง, Keep aPTT ratio 1.5-2.5

Heparin

Side effect

- Bleeding
- Thrombocytopenia : (HIT : heparin-thrombocytopenia)

HIT : ภาวะเกล็ดเลือดต่ำรุนแรง

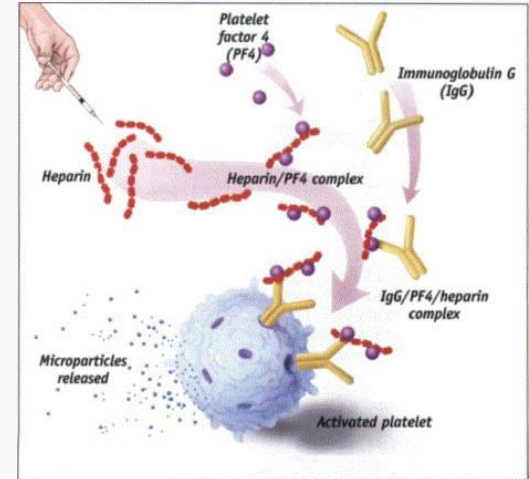
1.Type I : nonimmune -mediated reaction : ไม่รุนแรง หายได้เอง

2.Type II : immune-mediated reaction

(Platelet ลดต่ำ หลังจากเริ่มยา 5-10 วันหลังเริ่มยา

- Monitoring

Platelet ลดลง 50 % จาก base line





Enoxaparin: Dose & Monitoring

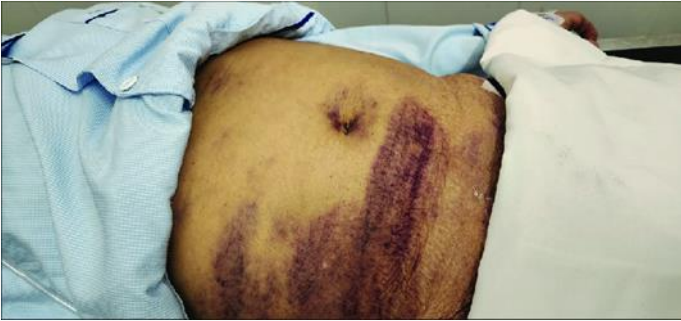
Standard dosing	Severe renal impairment Crcl < 30 mL/min
Treatment of ACS (Recommend duration : 2-8 days)	
1 mg/kg twice daily, SC	1 mg/kg once daily, SC
Treatment of acute STEMI with Fibrinolytic < 75 years	
30 mg IV bolus Plus 1 mg/kg SC (Total 2 dose max 100 mg) Followed 1 mg/kg SC q 12 hr.	30 mg IV bolus Plus 1 mg/kg SC (Total 2 dose max 100 mg) Followed 1 mg/kg SC q 24 hr.
Treatment of acute STEMI with Fibrinolytic ≥ 75 years	
0.75 mg/kg SC q 12 hr. (Total 2 dose max 75 mg)	1 mg/kg SC q 24 hr. (max 100 mg)



Monitoring : Level anti Xa : หลังให้ยา 4 ชั่วโมง (Peak Level)

Contraindication in Crcl < 15 mL/min

Enoxaparin-related hematoma



Fondaparinux

Indication :

- ❑ ACS : STEMI after fibrinolysis -> 2.5 mg sc q 24 hr.
- ❑ VT/PE :

BW < 50 kg. -----> 5 mg sc once daily

BW 50 -100 kg -----> 7.5 mg sc once daily

BW > 100 kg -----> 10 mg sc once daily

- Should be not support tp PCI >>> Risk catheter thromboembolism

- Contraindication in Crcl < 30 mL/min



Anticoagulant : Oral

□ Indication : AF,VTE

Vitamin K antagonist

warfarin

ชื่อการค้า: Maforan, Orfarin
ขนาดยา 2 mg, 3 mg, 5 mg



Direct Oral Anticoagulants

Dabigatran



Edoxaban

Rivaroxaban



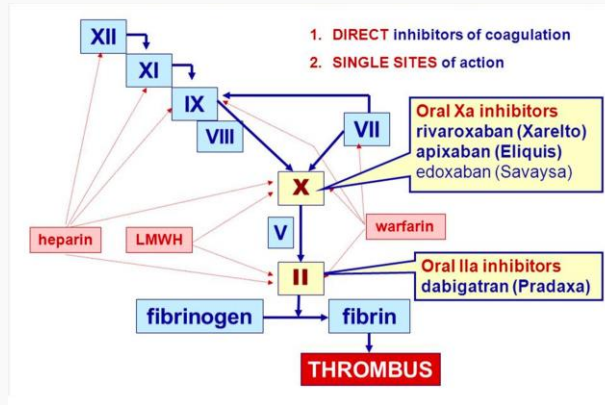
Apixaban

Warfarin



DOAC

Narrow therapeutic
Monitor : INR 2-3



- ❑ Rapid onset
- ❑ ไม่ต้อง monitor INR
- ❑ Drug interaction น้อยกว่า
- ❑ Avoid in severe renal impairment
- ❑ NOT USE : mechanical valve replacement, Moderate to severe Mitral stenosis

Drug interaction : Warfarin

INR



Amiodarone

NSAID : Ibuprofen Naproxen

Trimethopom - sulfamethoxazole

Quinolone : Ciprofloxacin, Levofloxacin

Macrolide : Azithromycin, Clarithromycin

อาหารเสริม / สมุนไพร : เปะก๊วย ฟ้าทะลายใจ กระเทียม น้ำมันปลา แครนเบอร์รี่ Grapefruit

Risk Bleed

INR



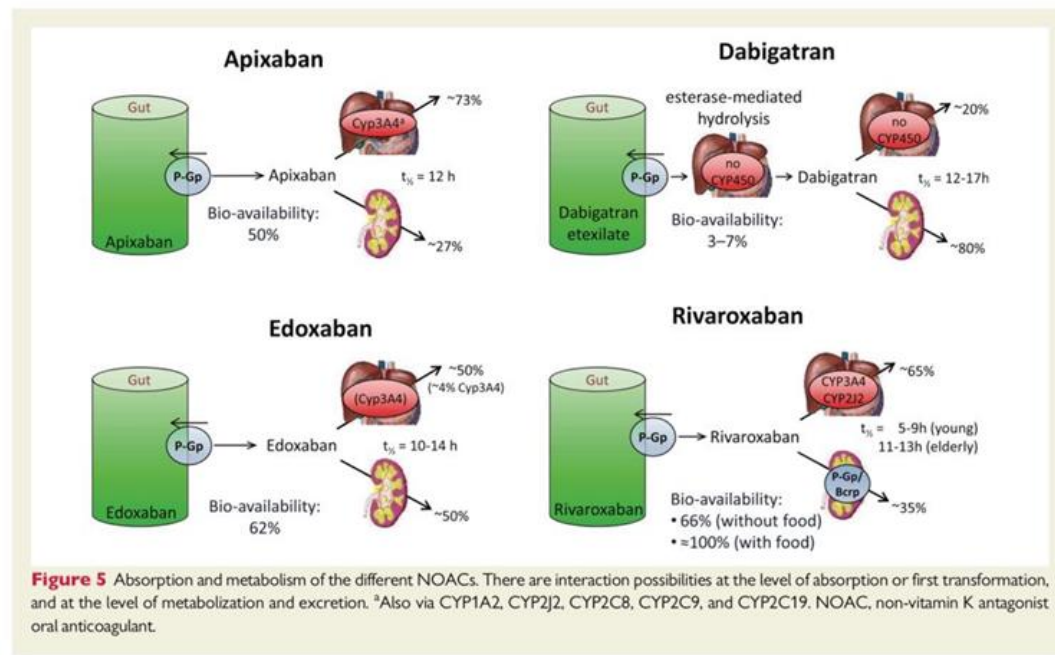
Rifampicin

Dicloxacillin

อาหารเสริม / สมุนไพร : ผักใบเขียว โสม ชาเขียว

Risk Embolism

Pharmacokinetic : DOAC

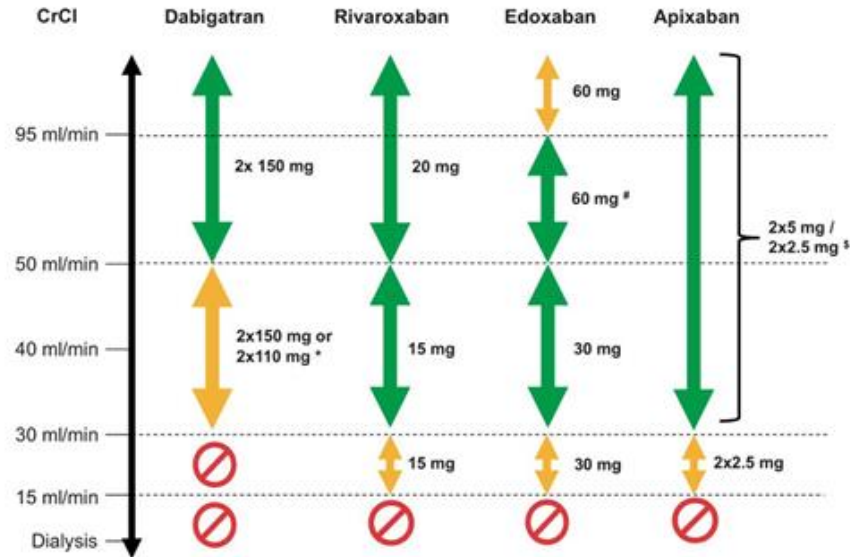


Direct Oral Anticoagulants

- ขนาดยา

	Stroke prevent in AF	Treatment DVT/PE		Long-term prevention of recurrent DVT/PE
	Standard dose	Initial therapy	Treatment phase	Standard dose
Dabigatran	110 mg BID/150 mg BID	Heparin/LMHW	110 mg BID/150 mg BID	110 mg BID/150 mg BID
Rivaroxaban	20 mg OD	15 mg BID, 21 days	20 mg OD	20 mg OD
Apixaban	5 mg BID	10 mg BID, 7 days	5 mg BID	5 mg BID
Edoxaban	60 mg OD	Heparin/LMHW	5 mg BID	60 mg OD

Renal dose adjustment



Apixaban

ในผู้ป่วยที่มี 2 ใน 3 ข้อ ต่อไปนี้

1. อายุ ≥ 80 ปี
2. น้ำหนัก ≤ 60 kg
3. Serum Creatinine ≥ 1.5 mg/dL

Drug interactions

ยาที่เพิ่มฤทธิ์
ยากลุ่ม DOAC



- ยาด้านเชื้อรา Ketoconazole, Itraconazole
- ยาด้านไวรัส : Ritonavir
- ยาด้านแบคทีเรีย : Erythromycin, Clarithromycin
- Anti-arrhythmic drug : Dronedarone, Amiodaron

ยาที่ลดฤทธิ์
ยากลุ่ม DOAC



- ยากันชัก : Phenytoin, Carbamazepine, Phenobarbital
- ยาวัณโรค : Rifampicin



Question ยาชนิดใดบ้างที่ไม่สามารถหัก แบ่ง/บด ได้ ?

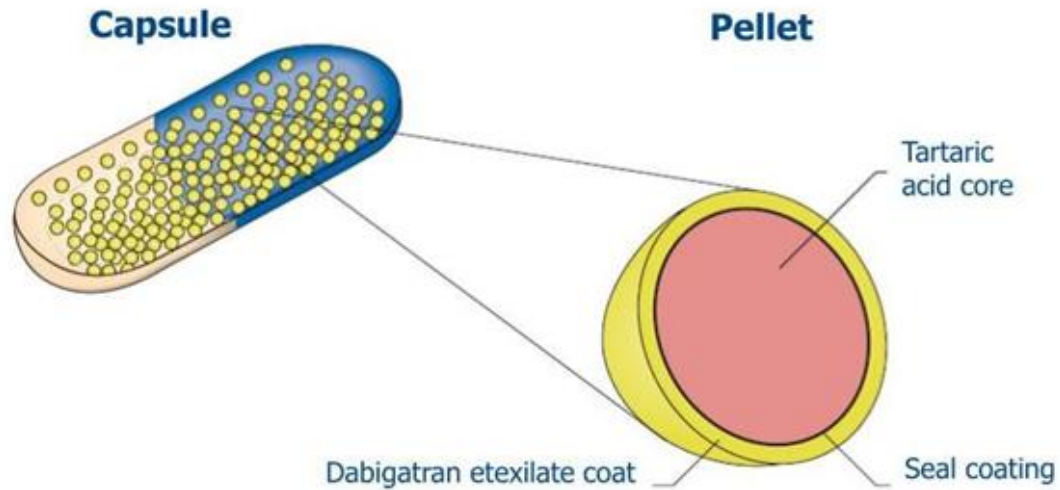
A.Dabigatran

B.Rivaroxaban

C. Edoxaban

D.Apixaban

Dabigatran





Question ยาชนิดใดต้องกินพร้อมอาหาร?

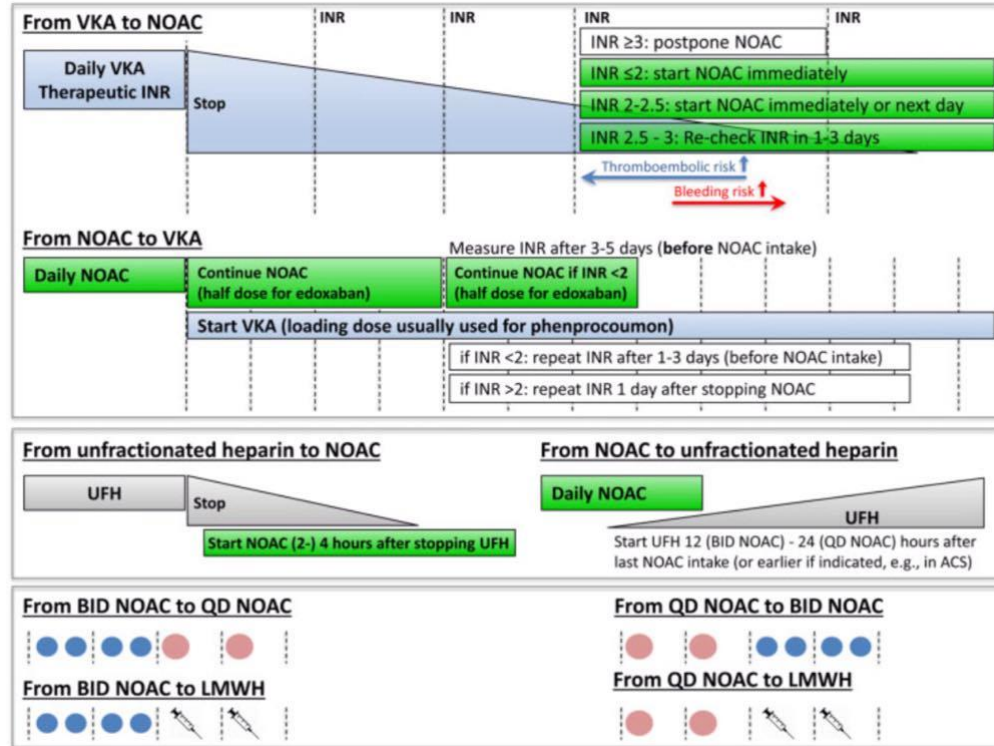
A.Dabigatran

B.Rivaroxaban

C. Edoxaban

D.Apixaban

Switching between NOAC and other anticoagulant





Question

ผู้ป่วยทาน Dabigatran 110 mg เวลา 06.00 น. และ 18.00 น.

แพทย์ต้องการเปลี่ยนไปใช้ Enoxaparin 0.6 ml. SC q 12 hr. ควรเริ่มฉีดตอนกี่โมง ?

(Last dose Dabigatran 18.00 น.)



Thanks you